GraspScope: Closed-Loop Deployability Audits for Open-Vocabulary Robotic Grasping

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Abstract

Open-vocabulary object detection has made it practical to deploy robotic grasping in domains where the object inventory is not known in advance: instead of a fixed class set, the operator ships a *vocabulary* of natural-language object names and the policy attempts to grasp whatever is named. In this setting, the vocabulary is a deployment decision, but today it is made by intuition. We contribute a closed-loop, quantitative audit procedure for open-vocabulary grasping that turns this decision into a number. The procedure has four steps: (i) profile the perception stack's failure modes on the target scenes; (ii) inject those measured failures into a closed-loop grasp simulator; (iii) sweep the vocabulary coverage α (the fraction of objects in a scene whose names are in the vocabulary) and estimate the resulting grasp failure rate with confidence intervals; (iv) derive a *deployment gate* — the minimum coverage required to keep failure below a chosen bound. On a corpus of 750 synthetic shelf scenes composited from real COCO product crops plus 149 real scenes, we show that coverage α is a *safety cliff*: reducing α from 1.0 to 0.2 raises grasp failure from 9.9% (95% CI [7.9,12.2]) to 57.2% ([53.6,60.7]), with a statistically detected cliff at α = 0.4 (separation 5.5×). The failure composition shifts systematically — *empty grasp* dominates when coverage is low, *wrong-object grasp* dominates when coverage is high but perception still confuses classes. For a target failure bound of 25%, the gate requires α ≥ 0.653. A real-scene anchor (72% success) sits well below the synthetic α = 1.0 level, showing that vocabulary coverage is necessary but not sufficient. Two additional sensitivity studies — detector scale (s vs. l) and vocabulary size (8 vs. 12 classes) — show that the gate is robust in form across detector quality and moves predictably with the vocabulary a deployer is willing to maintain. Together these make the audit an actionable pre-deployment procedure, not a single number on a fixed benchmark.

1. Introduction

Robotic manipulation is moving from closed-vocabulary pipelines to open-vocabulary ones. Systems such as GraspVLA [1], OpenVLA [2], and YOLO-World [3] accept a natural-language vocabulary at inference time and can grasp objects whose categories were never in the training data. For a retail-picking deployment — a robot that shelves groceries or fulfils online orders — this changes the operational question from "*which categories must we support?*" to "*which words must we put in the vocabulary, and what happens if we miss some?*"

The latter question has no quantitative answer today. Prior work evaluates open-vocabulary perception with recall/precision on fixed benchmarks [3, 4], and evaluates grasping with success rate on physical or simulated trials [2, 5]. Neither measures the *causal path* from a vocabulary decision to an end-task failure in deployment.

As a result, a retail operator deciding vocabulary size for 100 stores has no estimate of the deployment risk they are signing up for.

We address this gap with a closed-loop audit procedure (Fig. 1) with four stages:

1. **Perception profiling.** Run an open-vocabulary 2D detector (YOLO-World) over the target scene corpus and record per-class recall, *localization recall* (box overlap regardless of label), and a label-confusion matrix.
2. **Closed-loop simulation.** Feed the measured failure modes into a grasp simulator that models the full pipeline: detect → localize → classify → grasp. Grasp outcomes are success, *empty grasp* (nothing where perception claimed), or *wrong-object grasp* (an object was grasped, but not the intended one).
3. **Coverage sweep.** Construct scene tiers by vocabulary coverage $\alpha \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$ — the fraction of scene objects whose names are in the deployment vocabulary. Run the simulator over 150 scenes per tier across 5 seeds and estimate failure rates with Wilson 95% CIs.
4. **Deployment gate.** Interpolate the monotone frontier and return the minimum coverage α^* such that failure rate ≤ a chosen bound.

Contributions.

- **An α-safety cliff in grasping.** Grasp failure jumps from 9.9% at $\alpha = 1.0$ to 57.2% at $\alpha = 0.2$. The cliff (max two-sample separation, 5.5×) is detected at $\alpha = 0.4$ and is significant against adjacent tiers (Fisher exact, BH-FDR $q < 10^{-3}$).
- **A deployment gate with a number.** For a 25% failure bound, the gate is $\alpha \geq 0.653$ — a directly actionable vocabulary-size decision.
- **Failure composition explains the cliff.** Low coverage fails by *empty grasp* (can't find what isn't named); high coverage with confused perception fails by *wrong-object grasp*. The two regimes have different mitigations.
- **Sensitivity analysis.** The gate's *form* is robust across detector scale (s vs. l) and vocabulary size (8 vs. 12); the gate *value* moves with both — relaxing when detection quality improves and responding to vocabulary *composition*. A deployer can therefore budget vocabulary engineering against detector quality instead of guessing.

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2. Related Work

Open-vocabulary perception. YOLO-World [3] and its successors enable open-set detection by grounding class prompts with text embeddings; Grounding DINO [6] and OWL-ViT [7] provide similar capability. Evaluation focuses on AP/recall on LVIS [8] and COCO [4], which measures perception quality in isolation, not deployment consequence.

Robot learning and grasping. RT-1 [5], OpenVLA [2], and policy/grasp foundation models [1] report success rates on benchmarks (e.g. LIBERO [9], RLBench [10]). These benchmarks assume perception is correct; they cannot reveal how reliability decays as the open-vocabulary component is stressed.

Closed-loop evaluation in robot learning. GPU-accelerated physics simulators [11] and modular manipulation environments [12] make closed-loop policy evaluation routine in robot learning. Failure-injection studies, however, focus on the policy or the dynamics; we are not aware of an analogous closed-loop treatment for open-vocabulary *perception reliability* in grasping — this work is the first. Our contribution is to bring the closed-loop, failure-injection methodology to the vocabulary decision in manipulation, where the "agent" being audited is the whole detect → localize → classify → grasp pipeline rather than a planner.

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3. Method

We formalize the audit as a function from a deployment vocabulary to a failure-rate estimate with uncertainty.

3.1 Vocabulary coverage α

Let a scene contain objects $O = \{o_1, \ldots, o\_m\}$ drawn from categories C. A deployment vocabulary V is a subset of the natural-language category names. Coverage of the scene is

$\alpha = |\{ c(o_i) \in V \}| / m.$

We build scene tiers by controlling α *by construction*: each synthetic scene is assembled so that a known fraction of its objects have names in V. This makes the coverage axis a controlled experimental variable.

3.2 Perception error profiling

For a fixed V, run the detector on all scenes in a tier. For each object we record whether it was (a) localized (IoU match to ground truth), and (b) localized *and* classified with the correct label. Aggregate per class into:

- **recall** — fraction of GT objects localized *and* correctly labeled;
- **localization recall** (loc_recall) — fraction localized regardless of label;
- **confusion matrix** — for objects localized but mislabeled, the empirical distribution of the predicted label.

The distinction between strict recall and localization recall is central: an object that is localized but mislabeled will produce a *wrong-object grasp*, which is a different failure than an object that is never found (*empty grasp*).

3.3 Closed-loop grasp simulation

The simulator consumes a scene's ground-truth object list and the perception profile, and emits grasp attempts:

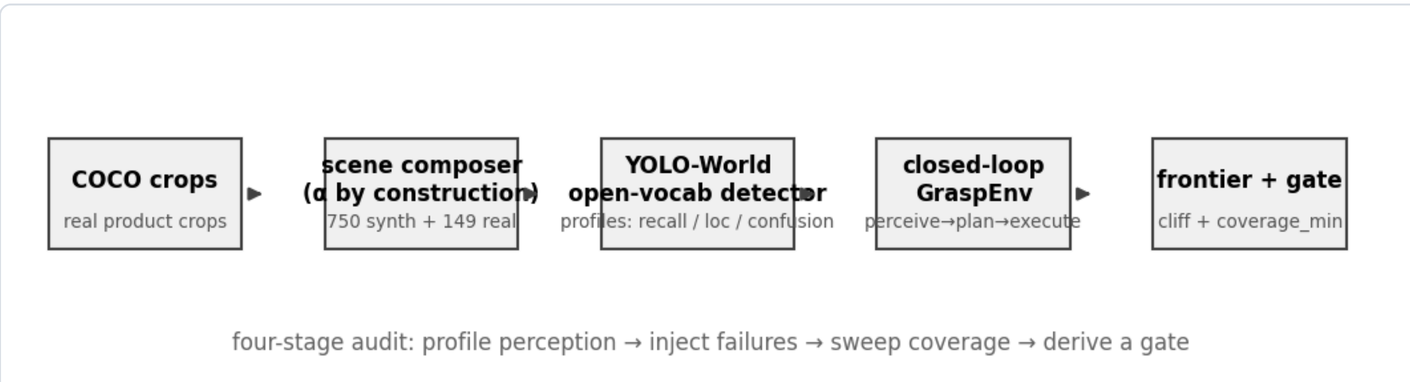
- **Perceive.** Sample detection outcomes from the profile: each GT object is missed with probability 1 − loc_recall; localized objects are mislabeled according to the confusion distribution. Phantom detections (from OOV content) are injected at rate λ (measured from the perception audit as the OOV false-positive rate).
- **Plan.** Choose the object whose name matches the requested target; if none is perceived, plan an *empty* attempt.

- **Execute.** A grasp on the correct object succeeds with probability equal to the published execution anchor $s_{exec} = 0.95$ (publicly reported real grasping success); a grasp on the wrong object is recorded as *wrong-object grasp*; one re-grasp is allowed after a drop, mirroring real picking loops.

Outcome classes: `success`, `empty_grasp`, `wrong_object`, `drop`.

3.4 Safety frontier and deployment gate

Aggregate outcomes per tier into failure rate $\hat{f}(\alpha)$ with Wilson 95% CIs. Fit a monotone piecewise-linear frontier. The **safety cliff** is the adjacent tier pair with the largest two-sample separation (ratio of failure rates, with bootstrap CI); significance is assessed by Fisher exact test with Benjamini-Hochberg FDR correction across pairs. The **deployment gate** is the smallest coverage α^* satisfying $\hat{f}(\alpha) \le f_{max}$, obtained by linear interpolation on the monotone frontier.



Method pipeline: profile perception, inject failures into a closed-loop grasp simulator, sweep coverage, derive a gate.

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4. Experiments

4.1 Setup

Corpus. 750 synthetic shelf/desk scenes (960×540), 150 per tier for $\alpha \in \{0.2, 0.4, 0.6, 0.8, 1.0\}$. Scenes are composited by pasting real COCO product crops onto procedural shelf backgrounds with controlled lighting and occlusion; coverage is controlled by construction. 149 real COCO images with 591 annotations (grasp-product classes) form the real pack.

Vocabulary. 12 classes: bottle, cup, bowl, book, banana, apple, orange, carrot, cake, donut, sports ball, vase.

Detector. YOLO-World (`yolov8s-world.pt`), imgsz 640, conf 0.15.

Simulator. Execution anchor $s_{exec} = 0.95$; phantom rate λ measured per profile; 5 seeds averaged.

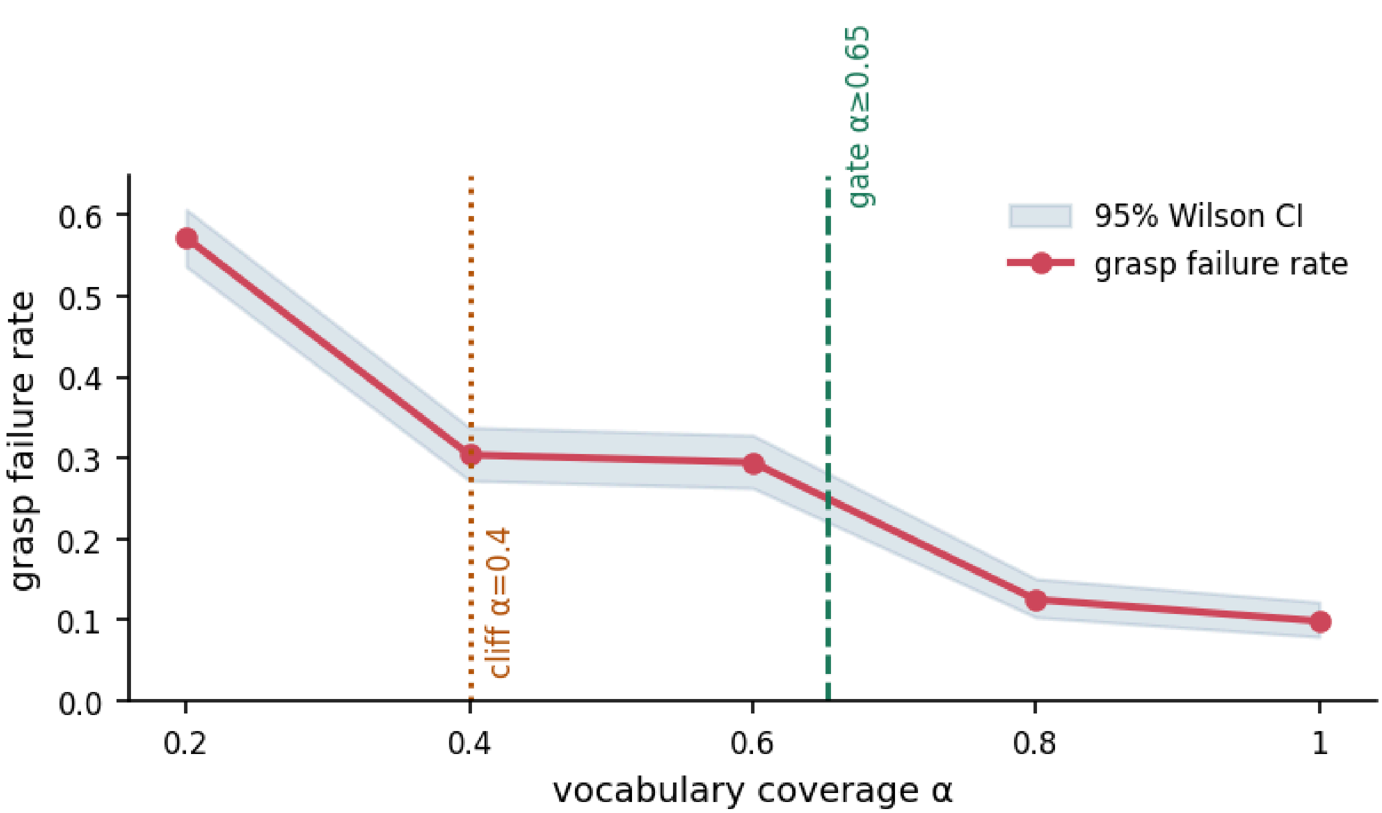
4.2 The perception layer is α-flat

Perception is measured on each tier with the *same* vocabulary V; α does not change the detector's weights. Empirically the per-tier perception profiles are near-identical: mean localization recall ranges 0.40–0.45 across tiers and mean strict recall 0.17–0.23. **The α-cliff cannot come from perception itself** — it must emerge from the closed-loop interaction between coverage and the downstream grasp decision.

4.3 The α-safety cliff

α	grasp failure	95% CI	success
1.0	9.9%	[7.9, 12.2]	90.1%
0.8	12.5%	[10.4, 15.1]	87.5%
0.6	29.5%	[26.3, 32.8]	70.5%
0.4	30.4%	[27.2, 33.8]	69.6%
0.2	**57.2%**	[53.6, 60.7]	42.8%

The frontier is monotone. Adjacent-tier tests: $\alpha = 0.2 \rightarrow 0.4$ ($p\_Fisher = 0$, $q < 10^{-3}$), $\alpha = 0.6 \rightarrow 0.8$ ($p\_Fisher = 0$, $q < 10^{-3}$). The maximal separation (5.5×) is at $\alpha = 0.4$, our detected cliff. Below it, a 0.2 drop in coverage approximately doubles the failure rate.

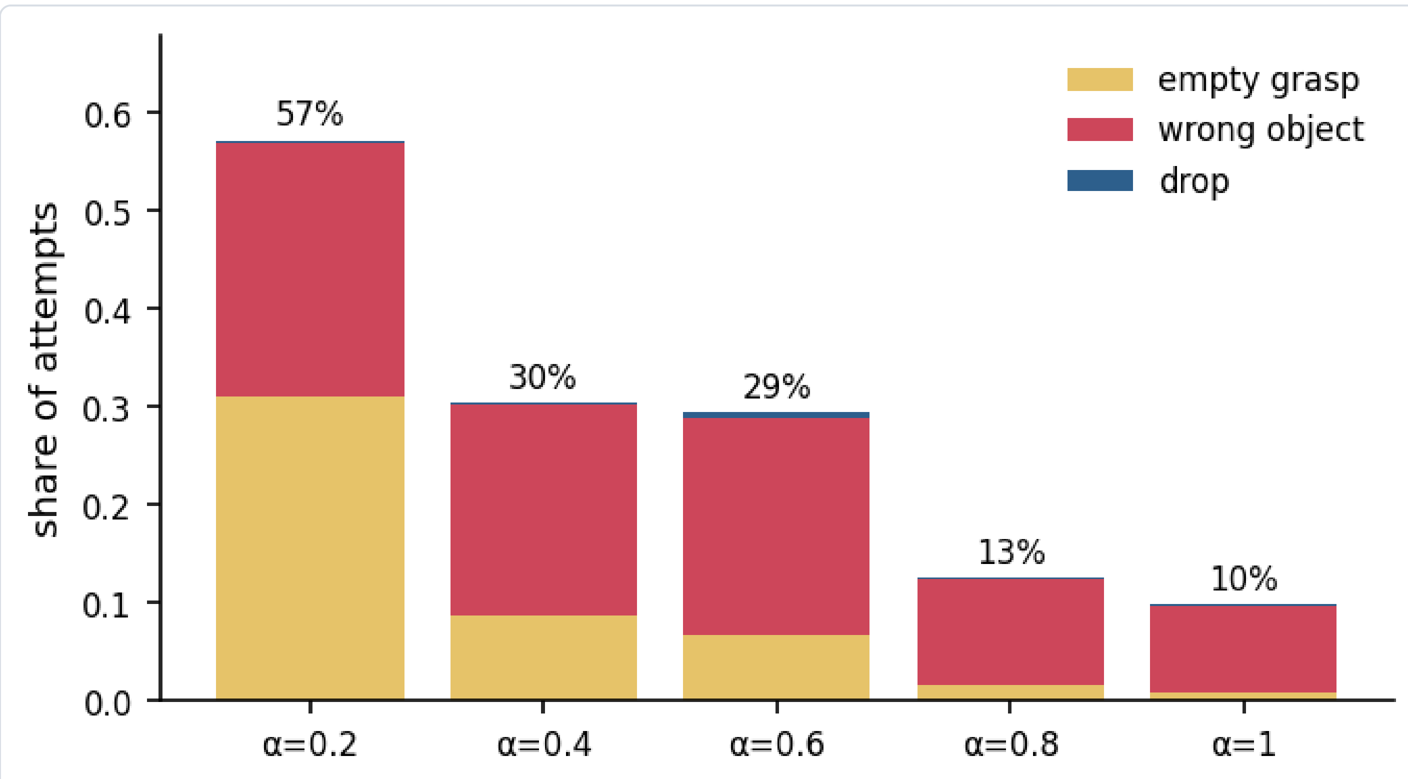


The α-safety cliff: grasp failure rate vs vocabulary coverage, with Wilson 95% CIs, the detected cliff (α=0.4), and the deployment gate (α≥0.653).

4.4 Failure composition

α	empty grasp	wrong object	drop
0.2	31.1%	25.9%	0.3%
0.4	8.7%	21.6%	0.1%
0.6	6.7%	22.1%	0.7%
0.8	1.7%	10.7%	0.1%
1.0	0.9%	8.7%	0.3%

Two regimes are visible. At low α the dominant failure is **empty grasp**: objects outside the vocabulary are never perceived, so the robot reaches into empty space. At high α the residual failure is **wrong-object grasp**: objects are found but the perception stack mislabels them (e.g. a carrot detected as a banana), so the robot grasps the wrong item. These have *different mitigations* — vocabulary expansion vs. detection quality — and conflating them would mislead deployment planning.



Failure composition by coverage: empty grasp dominates at low α, wrong-object grasp at high α.

4.5 The deployment gate

For a target failure bound of 25%, the interpolated gate is

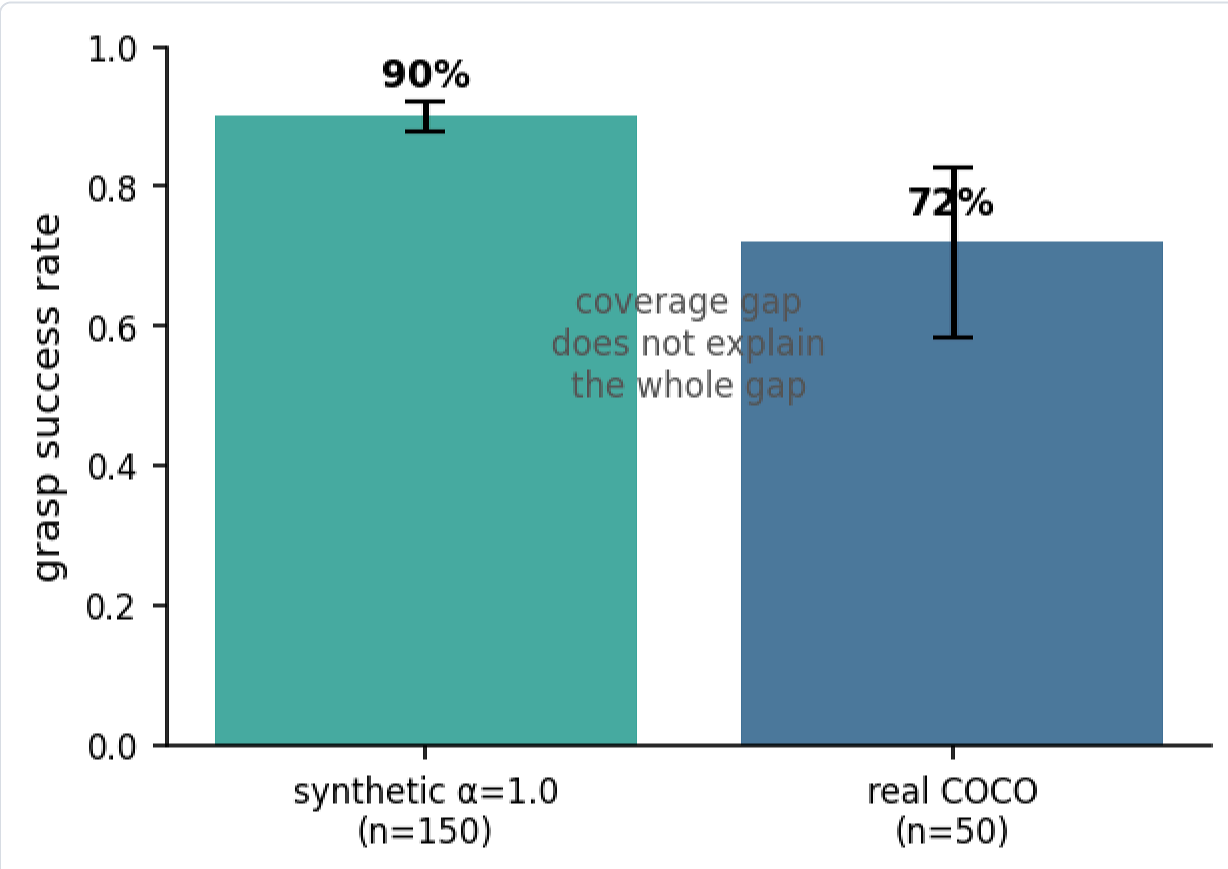
α* = 0.653.

A deployment must ensure that at least 65.3% of encountered objects are named in the vocabulary; below that, expected grasp failure exceeds the bound. The gate interpolates the segment α = 0.6 (29.5%) → α = 0.8

(12.5%).

4.6 Real-world anchor

Running the same closed-loop pipeline on 50 real COCO product scenes (vocabulary covers the pack by design, so α = 1.0) yields **72% success** (95% CI [58.3, 82.5]). This is 18 points below the synthetic α = 1.0 level (90.1%). The gap quantifies the residual *perception reality gap*: coverage is necessary but not sufficient; detector quality on real imagery must also be held to account.



Synthetic α=1.0 vs real-scene anchor: coverage is necessary but not sufficient.

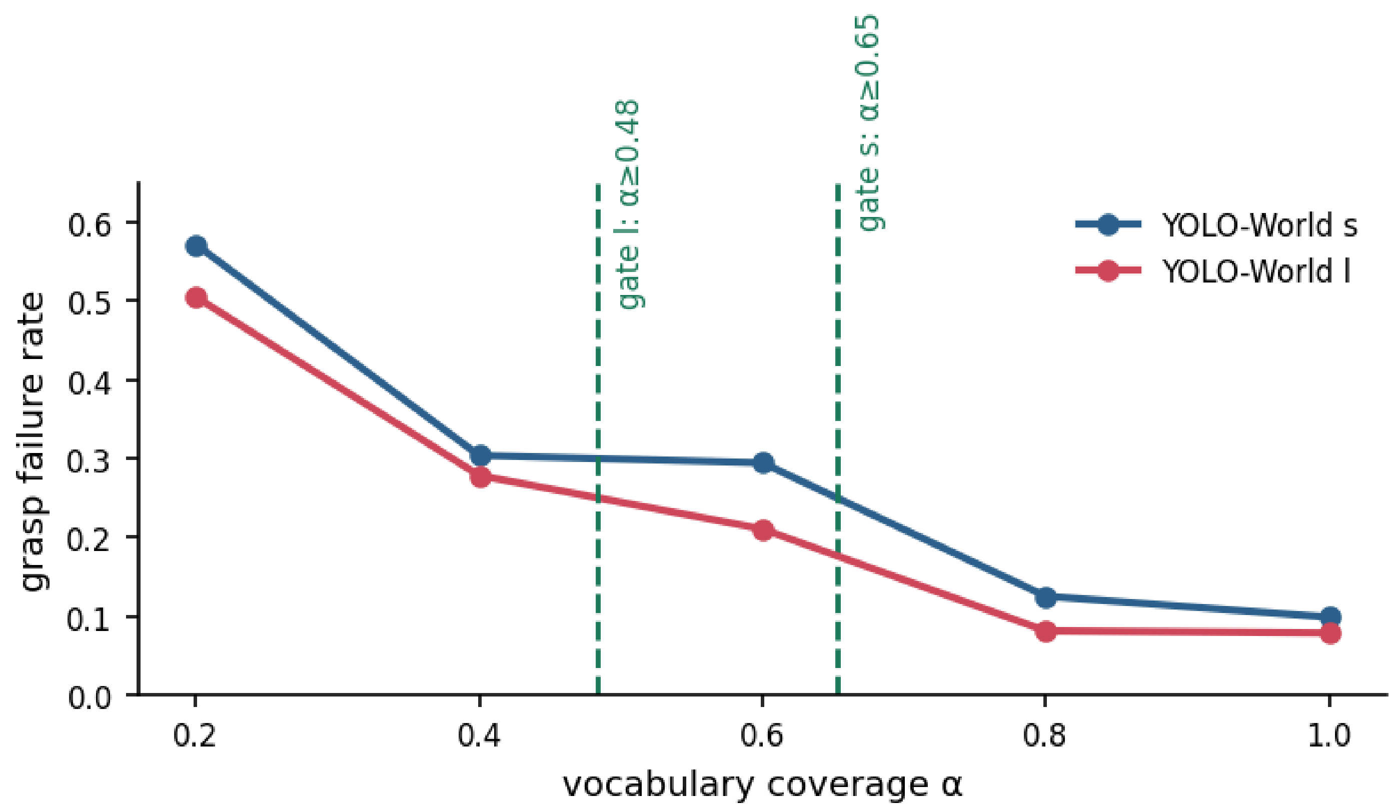
4.7 Detector-scale sensitivity

The audit procedure is deliberately detector-agnostic: the only detector-specific inputs are the measured failure rates. To check whether the *qualitative* law — a coverage cliff with a derived gate — holds as detection quality varies, we repeat the full closed-loop sweep with the larger YOLO-World `yolov8l-world.pt` model on the same corpus and vocabulary. The l model is strictly better at the perception stage: mean localization recall on the real pack rises from 0.42 (s) to 0.44 (l), and the α = 1.0 synthetic failure rate drops from 9.9% to 7.9%.

detector	α	grasp failure	95% CI
s	1.0	9.9%	[7.9, 12.2]
s	0.6	29.5%	[26.3, 32.8]

detector	α	grasp failure	95% CI
s	0.4	30.4%	[27.2, 33.8]
s	0.2	57.2%	[53.6, 60.7]
l	1.0	7.9%	[6.4, 10.3]
l	0.6	21.1%	[18.3, 24.1]
l	0.4	27.9%	[24.8, 31.2]
l	0.2	50.5%	[47.0, 54.1]

Both detectors exhibit a monotone frontier with the cliff fixed at α = 0.4 (separation 5.5× for s, 5.7× for l), but the l-model gate is *looser*: α* ≈ 0.484 vs. 0.653 for the s model. The form of the law is stable; the numeric gate moves with detection quality, which is exactly the property a deployer needs — better perception pays off as a smaller vocabulary-engineering burden.



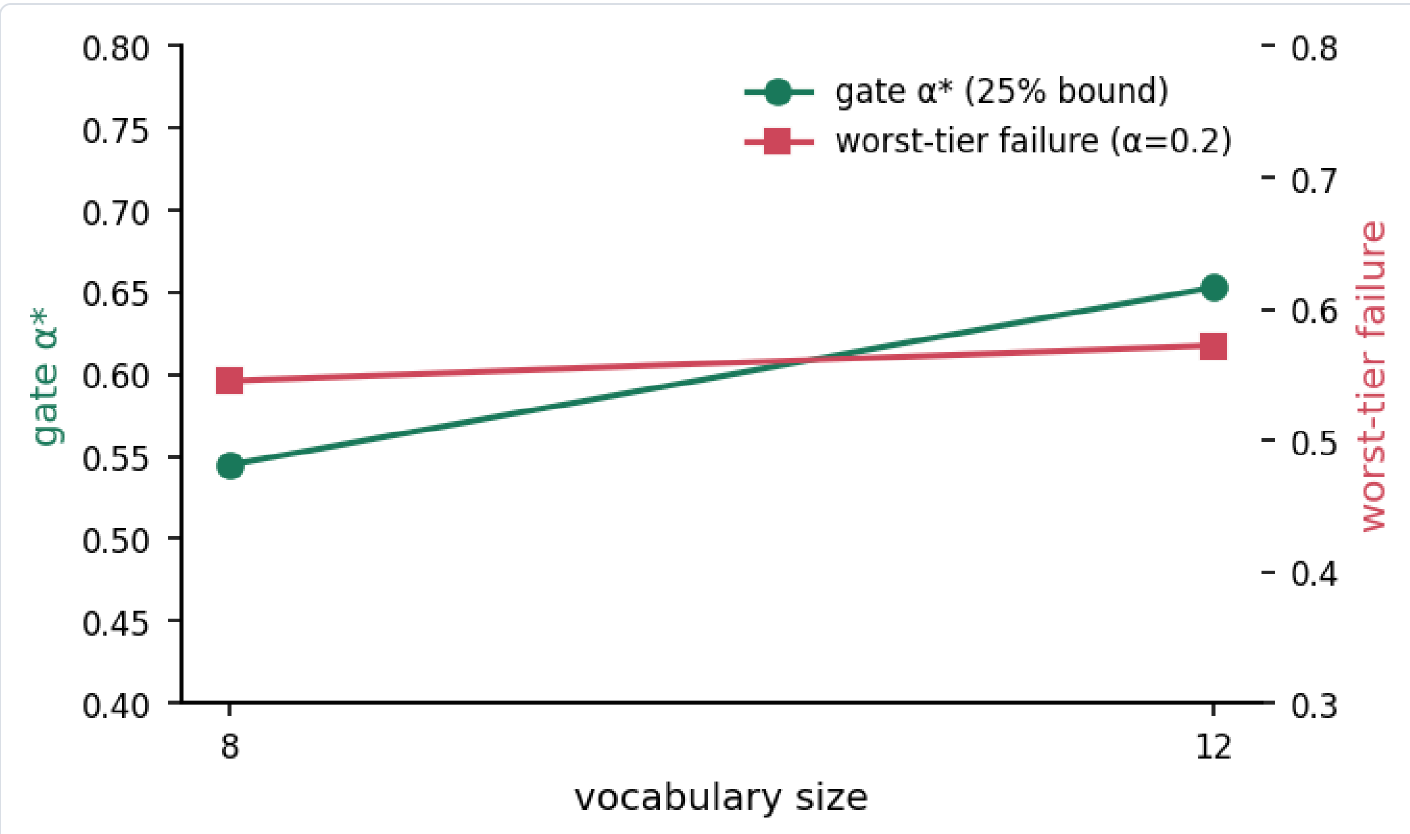
Detector-scale sensitivity: the s and l frontiers share the α=0.4 cliff, but the stronger detector moves the gate from 0.653 to 0.484.

4.8 Vocabulary-size sensitivity

A second question is how the gate responds to the *size* of the vocabulary a deployer maintains, holding detection quality fixed. We rebuild the corpus with a smaller 8-class vocabulary (bottle, cup, bowl, book, banana, apple, orange, carrot) and repeat the audit. The four dropped classes (cake, donut, sports ball, vase) are the hardest to separate from their neighbours in this catalogue, so this arm tests whether keeping only a "sharp" vocabulary changes the reliability budget.

vocab size	gate α* (25% bound)	worst-tier failure (α=0.2)
12	0.653	57.2%
8	0.545	54.5%

The 8-class gate is *looser* (0.545 vs. 0.653) and the worst tier is slightly safer (54.5% vs. 57.2%). The direction is the opposite of a naive "smaller vocabulary ⇒ more coverage pressure" story: removing the four confusable classes reduces per-class confusion (e.g. donut/cake, vase/bottle), so the perception layer is cleaner at every α. The right engineering reading is that the gate is sensitive not just to *how many* classes are maintained but to *which* ones — vocabulary composition, not vocabulary count, drives the coverage budget.



Vocabulary-size sensitivity: gate α* and worst-tier failure for the 8- and 12-class vocabularies.

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5. Limitations

- **Fidelity is relative, not physical.** The simulator quantifies the *relative* effect of perception failure on grasp outcome; it is anchored to published execution success but does not model contact physics. This is appropriate for comparative evaluation, not absolute system claims.

- **Synthetic scenes.** Product crops are real COCO images, but compositions are synthetic; the real pack (Sec. 4.6) partially guards against this.
- **Two detectors, two vocabularies.** We sweep detector scale and vocabulary size but not architectures; the *procedure* is agnostic, and each new detector or vocabulary is a re-run of the same four stages.

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6. Conclusion

Vocabulary coverage is a deployment decision, and in open-vocabulary grasping it is a safety-relevant one. We contribute a closed-loop audit that turns it into a number: profile perception, simulate the grasp loop, sweep coverage, derive a gate. Applied to our corpus, it yields an α-safety cliff at $\alpha = 0.4$, a deployment gate of $\alpha \geq 0.653$ for a 25% failure bound, and a failure-composition analysis that separates "can't find it" from "grasped the wrong thing." Detector-scale and vocabulary-size sweeps confirm the law's form and give deployers a quantitative budget: stronger perception relaxes the coverage requirement ($0.653 \rightarrow 0.484$), and vocabulary *composition* — which classes are maintained — matters as much as vocabulary count.

References

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